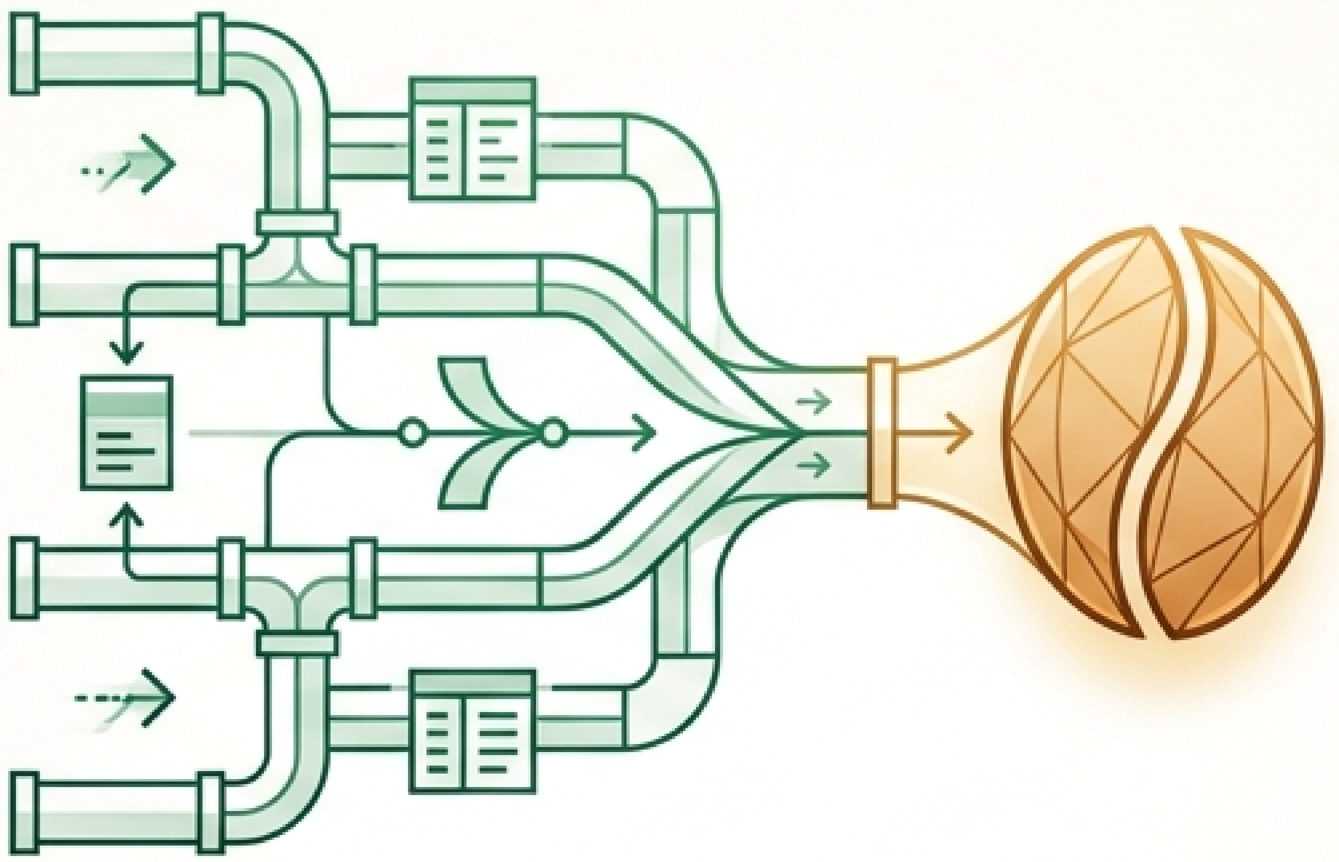


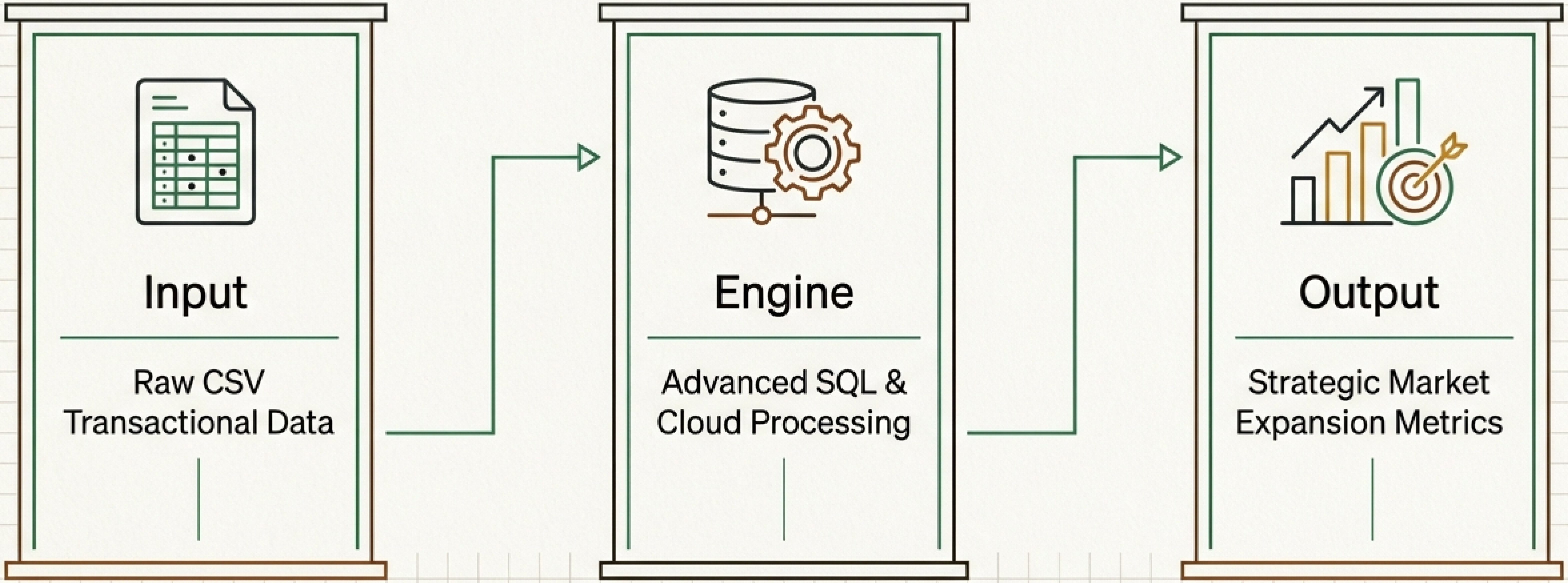
Coffee Sales Analysis

Translating raw transactional data into actionable business intelligence through advanced SQL.

End-to-End Project Showcase | BigQuery & MySQL

From Raw Transactions to Strategic Intelligence

This project architects an end-to-end data pipeline to analyze coffee sales data. By utilizing advanced SQL constructs across dual platforms (BigQuery and MySQL), the analysis cross-validates performance while extracting temporal trends, product profitability, and spatial customer dynamics.



Core Business Questions Addressed

What are the sales trends across daily, weekly, and monthly periods?

Which products and categories drive the highest revenue and margin?

How do purchasing patterns shift across distinct customer regions and segments?

Which customer cohorts represent the highest lifetime value (LTV)?

Cross-Platform Tech Stack

SQL

BigQuery

Deployed for large-scale data processing, cloud infrastructure integration, and query optimization.

The core extraction, transformation, and aggregation language linking both environments.

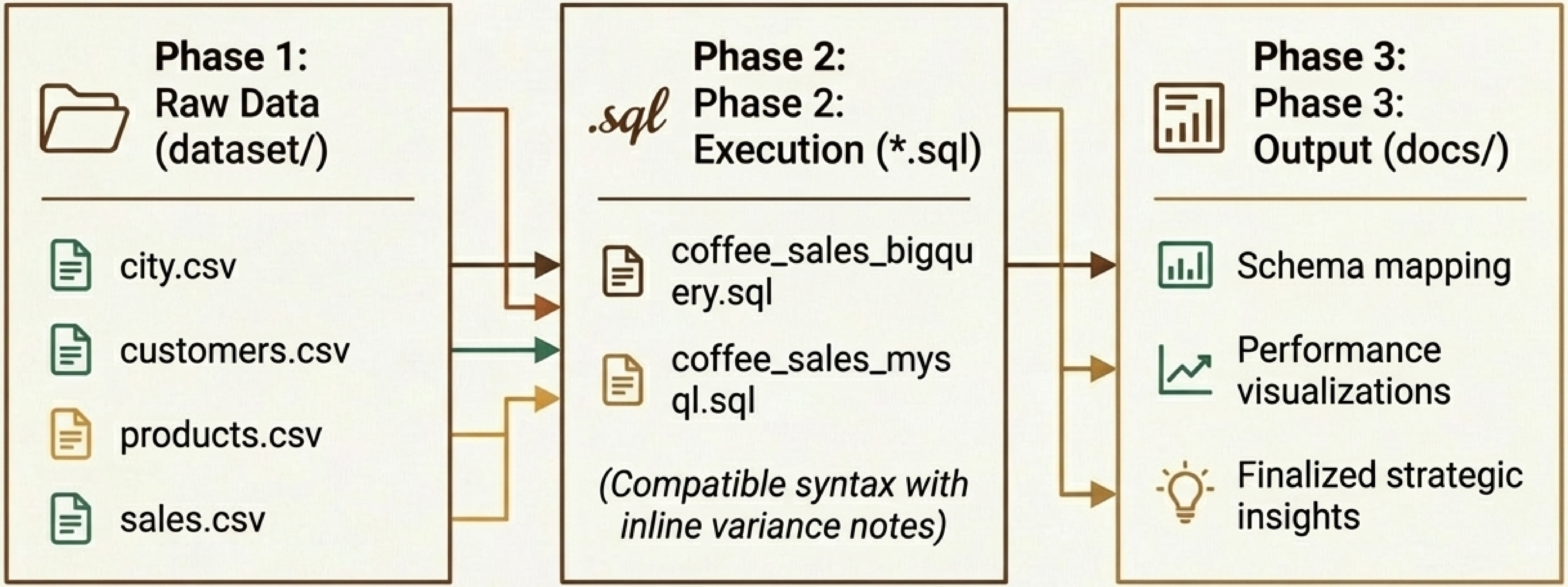
MySQL Workbench

Utilized for rigorous relational schema design, query development, and localized data validation.

Query Engineering & Techniques

Technique	Execution	Business Value
Window Functions	`ROW_NUMBER, RANK, DENSE_RANK with PARTITION BY / ORDER BY`	Generating running totals and moving averages to map temporal sales momentum.
Common Table Expressions (CTEs)	`Recursive CTEs and multi-step WITH clause chaining`	Untangling hierarchical data and optimizing complex query performance.
Aggregations & Joins	`Multi-table joins across customers, products, and transactions`	Calculating true group-level revenue, margin, and frequency metrics.

Project Structure & Workflow



The Data-Driven Expansion Strategy

“The ultimate goal of this analysis is to identify optimal locations for new store openings. The criteria demands a delicate balance of revenue potential, customer volume, and baseline operational cost efficiency.”

Outcome: Three target cities emerged as clear frontrunners.

01

Pune: Optimized Profitability

Pune offers the strongest unit economics among all top candidates, balancing high revenue with exceptionally low overhead per customer.

Revenue

Highest total revenue across all analyzed cities.

Cost Efficiency

Lowest average rent per customer among top candidates.

Behavior

Strong average sales per customer, indicating high individual buyer value.

02

Delhi: Unmatched Market Scale

Delhi is the volume play. It offers a massive addressable market with manageable costs, ideal for high-traffic footprint models.

Addressable Market

Largest estimated coffee consumer base (7.7 Million).

Current Traction

Highest total customer count (68 active cohorts).

Cost Cap

Average rent per customer remains manageable at ₹330, staying well under the ₹500 viability threshold.

03

Jaipur: High Yield, Low Overhead

Jaipur represents a highly efficient expansion target, blending a large existing customer base with incredibly low operational barriers.

Volume

Highest raw number of engaged customers (69).

Cost Efficiency

Drastically low rent per customer (₹156).

Value

Solid average sales per customer at ₹11.6k, creating a highly favorable LTV-to-CAC equivalent ratio.

City Recommendation Matrix

Target City	Primary Advantage	Customer Base	Avg Rent/Customer	Sales/Customer
Pune	Unit Economics	Moderate	Lowest	Strong
Delhi	Market Scale	Massive (7.7M)	₹330 (Manageable)	Moderate
Jaipur	Efficiency	High (69 active)	₹156 (Very Low)	₹11.6k

Takeaway: Lower rent per customer combined with higher average sales isolates the strongest unit economics for rapid ROI on new physical locations.

Explore the Full Analysis

Comprehensive SQL queries, schema diagrams, and data dictionaries are available in the repository. All queries are fully optimized for both BigQuery and MySQL 8.0+.

github.com/kalash-thakre/Coffee-Sales-Analysis

Designed for data-driven decision making.